

An all-pairs catalogue of real three-dimensional close encounters between numbered main-belt asteroids during the Gaia DR3 window, with a measured incompleteness budget

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ABSTRACT

Context. Close approaches between asteroids are the classical route to asteroid mass determination and a dynamical observable of the main belt. Systematic encounter searches exist — from early surveys (Galád & Gray 2002) to mutual-encounter catalogues built for Gaia astrometry (Ivantsov et al. 2018) and the signal-ranked search behind the largest mass catalogue to date (Fuentes-Muñoz et al. 2025) — but their event lists are published truncated (top candidates per perturber) or without a characterised selection function: no all-pairs 3D encounter catalogue has been published in full with provenance and a *measured* incompleteness budget.

Aims. We build a systematic catalogue of real 3D close encounters between numbered main-belt asteroids (semi-major axis 1.5–4.0 AU) during the Gaia DR3 window (2014 July 25 – 2017 May 28), publish it in full, and, unlike a bulk propagation exercise, we *measure* what the detection pipeline misses rather than asserting completeness.

Methods. Orbits are propagated from a single frozen MPCORB snapshot centred on the window. Encounters are found by a per-timestep 3D KD-tree spatial query and refined on a dense temporal sub-grid. Reported distances use an analytic two-body refiner; a hybrid variant carries full N-body (rebound/ASSIST) refinement for the dynamically fragile (low-perihelion or high-eccentricity) subset. Perturber masses come from a joint mass+orbit least-squares fit to Gaia FPR astrometry.

Results. The catalogue records the minimum physical 3D separation — a genuine spatial approach, not a sky-plane co-location — for 80 072 774 encounter pairs within 0.05 AU, with encounter epoch, relative velocity, geometry, and estimated physical properties. Its distinguishing feature is a *measured* completeness budget: the Kepler-to-N-body refinement error (median 12 μ AU, p99 2.5 mAU) and the threshold-induced false-negative rate (a 0.70 % downward-crossing rate within the discarded 0.05–0.06 AU band, symmetric and scatter-dominated, implying ~ 0.25 % of the catalogue censored) bound this catalogue directly, while the recall of an orbital prefilter on the adverse tail (76 %) is quantified as a caveat for prefiltered runs. Applied to mass determination, the engine recovers the four calibrators at $|z| < 3$ (Ceres, Vesta and Hygiea as measurements; Pallas, with only eight encounters, as a bound) and fits sixteen perturbers, six of them identifiable; these agree at $|z| < 2$ with recent independent determinations — Fuentes-Muñoz et al. (2025) on the same Gaia FPR data and Siltala & Granvik (2022) on independent Gaia DR2 plus ground-based astrometry — a consistency check, not a proof of accuracy.

Conclusions. The catalogue is a reusable, defensibly characterised target-selection resource for asteroid dynamics and mass determination. It is explicitly not complete; the large-large ($D \gtrsim 100$ km, measured) approaches it contains all involve well-studied bodies, and no unpublished large-body mass emerged from the numbered population.

Key words. minor planets, asteroids: general – celestial mechanics – astrometry – catalogs – surveys – methods: data analysis

1. Introduction

Close encounters between asteroids are of interest on several fronts. A slow, close approach between a massive perturber and a small test body deflects the latter’s orbit measurably, and such events are the classical route to asteroid mass determination (Michalak 2000; Fienga et al. 2003; Goffin 2014; Baer & Chesley 2017; Siltala & Granvik 2020; Li et al. 2023; Fuentes-Muñoz et al. 2025). Encounters between members of the same dynamical family probe the collisional history of the belt, and the statistical distribution of mutual approach distances and velocities is itself a dynamical observable.

Systematic encounter searches have a long history. Galád & Gray (2002) searched for mass-determination encounters across 24 599 asteroids; Ivantsov et al. (2018) built a catalogue of mutual encounters of the numbered population over 2013–2023 specifically to exploit Gaia astrometry; Goffin (2014)

fitted masses against the entire numbered population rather than curated target lists; and Fuentes-Muñoz et al. (2025) ran a systematic, signal-ranked search over 1 783 perturbers against more than a million orbits — using, for main-belt targets, the same 0.05 AU threshold adopted here. What none of these works provides is the *event list published in full with a characterised selection function*: published tables are truncated to the top candidates per perturber, and the completeness of the underlying searches is asserted rather than measured. That is the gap this work fills: an *all-pairs* catalogue over the numbered main-belt population ($1.5 \leq a \leq 4.0$ AU) in the Gaia DR3 window, published in full with provenance, where — the point that separates this catalogue from a bulk propagation exercise — we *measure* what the detection pipeline misses rather than asserting completeness.

A methodological distinction is central. Two asteroids can appear arbitrarily close in right ascension and declination as seen from Earth while remaining astronomical units apart along the

line of sight. This catalogue detects the former case only when it is also the latter: it records the *minimum physical separation in three dimensions* between the two bodies' heliocentric positions, independent of any observer. It is therefore a catalogue of genuine spatial approaches, not of sky-plane co-localisations. Light-time correction — essential when comparing model positions to Gaia's astrometry — is deliberately *not* applied here, because the encounter is a purely geometric relation between two propagated trajectories at a common dynamical time.

The remainder of this paper describes the data and detection method (Sect. 2), the measured completeness budget that is the catalogue's distinguishing feature (Sect. 3), the notable individual events surfaced by mining the catalogue (Sect. 4), the application to mass determination (Sect. 5), data availability (Sect. 6), and conclusions (Sect. 7).

2. Data and method

2.1. Sources and conventions

Orbital elements are taken from a single frozen MPCORB snapshot, `MPCORB_20160217.DAT` (SHA-256 prefix `3e44e7d3...`), selected because its osculating epoch lies at the centre of the Gaia DR3 window, minimising the propagation distance — and hence the accumulated two-body error — for Kepler refinement. The snapshot hash and download date are recorded in the catalogue's provenance sidecar so that results are traceable to an exact input; MPCORB is a living file and results are not bit-reproducible across snapshot versions. The run is restricted to *numbered asteroids* (higher orbital quality); provisional designations are out of scope for this release. The sample is further cut to the main belt by semi-major axis, $1.5 \leq a \leq 4.0$ AU, which excludes near-Earth asteroids ($a < 1.5$ AU), the Jupiter Trojans, and the outer/trans-Jovian populations ($a > 4$ AU). The resulting universe is $N = 449,454$ propagated bodies; this N is the population denominator for every completeness and censoring figure below, and is recorded in the provenance sidecar. “Complete” therefore always means complete *with respect to this numbered main-belt sample*, never the whole catalogue of known asteroids.

The temporal domain brackets the Gaia DR3 Solar System survey period (Tanga et al. 2023): the propagation window is 2014 July 25 – 2017 May 28, whose start (the DR3 astrometric-resolution baseline) slightly precedes the 2014 August 5 start of nominal SSO operations, so the window fully contains the survey coverage. The propagation grid extends modestly beyond each nominal edge (≤ 2 d) so that minima near the boundary are bracketed rather than clamped; an encounter whose refined minimum nonetheless falls outside 2014 July 25 – 2017 May 28 carries an `edge_censored` flag and is excluded from the tabulated showcases (Sect. 4). All internal computation is carried out in Julian Date on the Barycentric Dynamical Time (TDB) scale; conversions to/from the Barycentric Coordinate Time (TCB) that Gaia reports, and to UTC, are performed only at input/output interfaces. Distances are in AU, angles in radians, and velocities in AU/day internally, converted to user-facing units (km, degrees, km s^{-1}) only for output. Positions are computed in the barycentric ecliptic frame for the N-body coarse scan and in the heliocentric ecliptic frame for the Kepler refinement; because the catalogue records the *mutual* separation of two bodies, the common Sun–barycentre offset cancels in the difference vector and the recorded distances are identical in either frame. (The provenance sidecar's `reference_frame` field refers to the N-body scan.)

2.2. Detection

The naive cost is prohibitive: the 449 454 numbered bodies over a multi-year window at hour-level cadence imply of order 10^{11} pairwise distances per step. Detection is therefore layered.

An *orbital prefilter* is available to prune pairs that cannot physically approach — the shipped heuristic keeps a pair only if $|\Delta a| \leq 0.5$ AU and $|\Delta i| \leq 30^\circ$. This filter is a deterministic mask on osculating elements. We note here, and quantify in Sect. 3.3, that at main-belt scale ($\gg 5000$ bodies) the pair-list prefilter is not built; the frozen run therefore applied only the KD-tree spatial query, and the $|\Delta a|$ cut did *not* shape this catalogue (provenance `prefilter.effective = "skipped_large_n"`).

The core of the detector is a *per-timestep 3D KD-tree*. At each coarse step the heliocentric ecliptic positions of all bodies are inserted into a KD-tree ($O(N \log N)$ construction), and a radius query returns neighbour candidates ($O(\log N)$ per query), avoiding any $O(N^2)$ distance matrix. The coarse cadence is $\Delta t = 12$ h. To prevent the cadence from missing an encounter whose true minimum falls *between* two samples, the query radius is *widened* to $r_q = 0.05 \text{ AU} + v_{\max} \Delta t \approx 0.0572 \text{ AU}$ with $v_{\max} = 25 \text{ km s}^{-1}$: under near-linear relative motion a pair whose separation dips below 0.05 AU between samples is within r_q at the nearer bracketing sample and therefore enters the candidate set; the sub-grid refinement then discards any whose true minimum exceeds 0.05 AU. Bracketing a minimum midway between two samples strictly requires only $v_{\max} \Delta t/2 \approx 0.0536 \text{ AU}$; the pipeline uses the full-step widening ($v_{\max} \Delta t$), a factor-of-two conservative margin. The bracketing is thus exact for relative velocities up to $v_{\max}$, and the residual (encounters with $v_{\text{rel}} > v_{\max}$) is bounded in Sect. 3.3.

Each coarse candidate is then *refined on a dense temporal sub-grid*. The coarse scan samples every $\Delta t = 12$ h, so a true minimum can fall up to $\Delta t/2 = 6$ h from the nearest coarse sample; the refinement window is therefore $\pm \Delta t/2 = \pm 6$ h around the bracketing sample — wide enough to contain the minimum for any pair — sampled at 120 s. Should the discrete minimum land on a window edge (the true minimum lying just beyond it), the window is re-centred on that edge and re-searched, up to four times, until the minimum is interior; a three-point quadratic vertex through the bracketing samples then yields the sub-sample epoch, minimum separation, and relative velocity. Time blocks are independent, so the scan parallelises across processes, and each pair's global minimum is deduplicated across block boundaries. This $\pm \Delta t/2$ window with iterative re-centring is the *refinement-window fix*: the pre-2026-07 catalogue used a ± 2 h window that, being narrower than $\Delta t/2$, failed to bracket the true minimum in roughly two-thirds of cases and truncated the affected minima at the window edge — a bias whose removal is confirmed by the injection–recovery regression test of Sect. 3.1.

Propagation is two-tiered. The coarse KD-tree scan is driven by an N-body rebound integration (WHFast; Sun + Jupiter + Saturn; $dt = 1$ h). The sub-grid refinement that produces the *reported* minimum distance runs through an analytic two-body Kepler propagator. Over the short Gaia DR3 arc (~ 3 yr from the central epoch) two-body propagation is adequate for the bulk of the belt, and its residual error is measured, not assumed (Sect. 3.1). For the dynamically fragile subset — pairs with perihelion $q_{\min} < 1.8$ AU or eccentricity $e_{\max} > 0.3$, 10 339 493 pairs (12.91 % of the catalogue) — a *hybrid* catalogue replaces the Kepler distances with full N-body refinement (rebound IAS15; Sun + 8 planets + 4 major asteroids; ± 12 h around the Kepler minimum; 60 s sampling). A pair whose N-body minimum reaches the outer 5 % of this win-

row is flagged `near_boundary` (the N-body refiner flags rather than re-centres), and the flag is retained so such rows can be held back from geometry-critical use. The hybrid catalogue carries refinement_method $\in \{\text{kepler}, \text{nbody}\}$ per row together with the Kepler and N-body values and their residuals, so users needing sub-mAU geometry on high- e / NEA-crossing pairs can cite it directly. Universal N-body refinement of all 80 M pairs (~ 400 CPU-days) is explicitly out of scope: the median error outside the fragile subset is already below 1 mAU (Sect. 3.1), so the marginal gain does not justify the cost.

Membership rule (mixed propagators). Catalogue membership is defined by the *Kepler* distance < 0.05 AU: a pair enters if its two-body refinement places it below threshold. In the hybrid product, therefore, the reported `dist_au` mixes propagators (two-body for the bulk, N-body for the fragile subset), and a small fraction of hybrid rows have an N-body distance slightly above 0.05 AU (the upward threshold crossings of Sect. 3.2). Analyses that require a homogeneous selection function should filter on a single propagator — either use the Kepler `dist_au` throughout, or restrict to `refinement_method = nbody` — rather than mixing the two.

2.3. Characterisation

Each pair contributes a single catalogue row: the *global* minimum separation of that pair over the observation window, with its exact epoch (from the refined sub-grid), the relative velocity at closest approach, and the geometry relative to Gaia. Secondary local minima of a pair that recurs within the window are not catalogued separately. A body is flagged *Gaia-observable* at the encounter when its solar elongation — computed geocentrically, which for Gaia at L2 differs by $\leq 0.6^\circ$ — exceeds 45° (outside Gaia’s solar-exclusion zone) and its apparent V magnitude — used as a proxy for Gaia’s G band, from which it differs by ≤ 0.2 mag for asteroids — is brighter than 21 (the faint limit); the pair-level flag is the logical OR of the two bodies. Of the 80 072 774 encounters, 15 251 039 (19.0 %) satisfy this for at least one body.

Physical properties are attached to both bodies; within each pair the two are ordered so that subscript 1 is the larger (by diameter) — the perturber, in the mass-determination sense. Diameters use direct measurements (IRAS, AKARI and NEOWISE thermal-infrared radiometry and stellar occultations; Tedesco et al. 2002; Usui et al. 2011; Mainzer et al. 2011, via the JPL Small-Body Database) where available; otherwise they are derived from the absolute magnitude H with an albedo taken — in decreasing order of preference — from a measured albedo, an orbital-zone average, or a default 0.14. A per-row provenance column records which of these applied, so H -derived (order-of-magnitude) sizes are distinguishable from measured ones in any size-based selection. The accompanying *class* is the *dynamical* class from the orbital elements (NEA/MBA/Trojan/Centaur/TNO), not a taxonomic (spectral) type.

The two encountering bodies are treated as independent test particles moving in the planetary + massive-asteroid field; their *mutual* gravitational deflection over the approach is not modelled. At asteroid masses and the $\geq 10^5$ km separations that dominate the catalogue this is entirely negligible for the recorded geometry, and it is in any case the very quantity a downstream mass determination sets out to measure, not a property of the encounter itself.

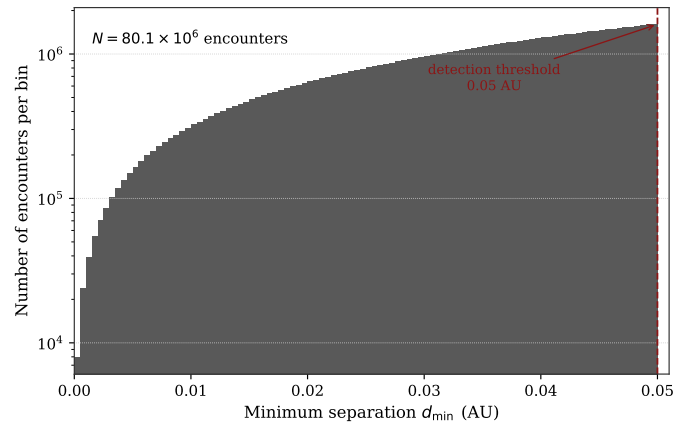


Fig. 1. Distribution of minimum 3D separations for the 80.1×10^6 catalogued encounters (log count per bin). The dashed line marks the 0.05 AU detection threshold.

2.4. Catalogue overview

The parent population and the encounter statistics are summarised in Figs. 1–3. The minimum-separation distribution (Fig. 1) rises toward the 0.05 AU threshold roughly as $dN \propto d^{1.0}$ (log-log histogram slope 1.01), so the cumulative count grows as $N(< d) \propto d^2$ — exactly the exponent expected for a uniform-density planar impact-parameter process (annulus area $\propto d^2$). The absence of any roll-over as $d \rightarrow 0.05$ AU is a free censoring diagnostic: a selection function that truncated encounters near the threshold would depress the slope in the outermost bin, and none is seen. The relative-velocity distribution (Fig. 2) peaks near its median of 4.28 km s^{-1} , typical of main-belt crossing velocities. The catalogue involves 449 213 distinct encountering bodies; their orbital distribution (Fig. 3), plotted from the frozen MPCORB elements with full coverage, reproduces the belt’s known structure — the 3:1, 5:2 and 2:1 Kirkwood gaps appear as clear voids — confirming the sample is the ordinary numbered main-belt population, not a biased subset.

Ranking by mass-perturbation signal. Sheer row count is not utility: the overwhelming majority of the catalogued pairs are sub-km $\times$ sub-km approaches with no measurable dynamical signal. To let users select the subset that matters — as signal-ranked searches such as Ivantsov et al. (2018) and Fuentes-Muñoz et al. (2025) do — each row carries a deflection-signal estimate, the impulse-approximation velocity kick $\Delta v \approx 2GM_1/(b v_{\text{rel}})$ imparted by the larger body (deflection_dv_m_s; GM_1 from the measured diameter and a zone-average density from Carry 2012). It is a ranking proxy, not a measurement. Ranked by it, the catalogue is steeply top-heavy: only 781 pairs exceed 1 mm s^{-1} and ~ 4000 exceed 0.1 mm s^{-1} , and the top of the list is dominated by (1) Ceres and (4) Vesta in slow, close passes — precisely the classical mass-determination configuration. This column turns the 80 M-row bulk into a targetable resource; the mass-determination application of Sect. 5 draws its test bodies from its high-signal tail.

3. Completeness budget

The value of this catalogue is not that it detects encounters — that is a bulk propagation — but that it states, with measured numbers, what it *misses* and how large the Kepler-vs-N-body distortion is. We stress the scope: the three *measured* terms be-

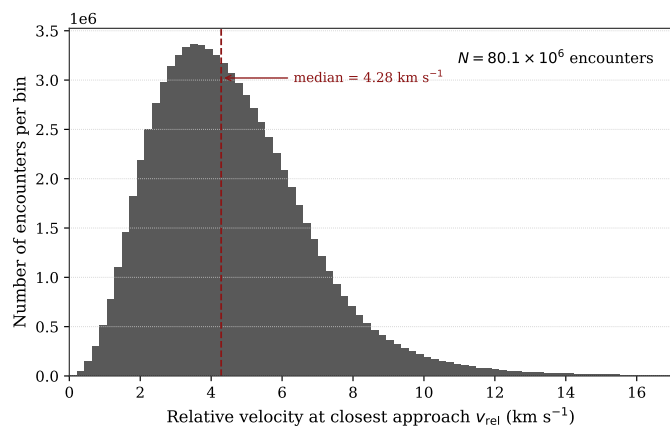


Fig. 2. Distribution of relative velocity at closest approach; the median is 4.28 km s^{-1} .

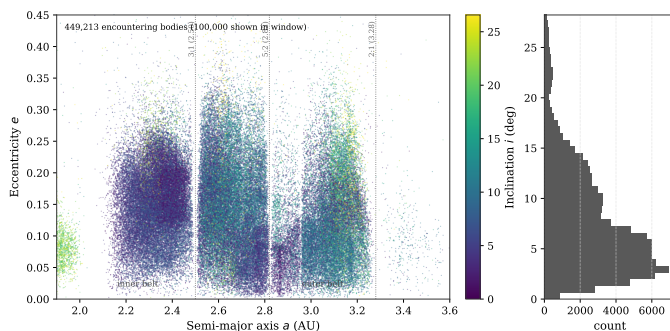


Fig. 3. Semimajor axis vs eccentricity of the encountering bodies, coloured by inclination, with the inclination marginal at right. Of the 449 213 distinct encountering bodies, osculating elements are available (and plotted, within the window shown) for 92 971; the remainder — mostly high-number, recently numbered objects absent from the orbital-elements source — are not shown. The omitted subset is *systematically* distinct (fainter, higher-number, and skewed toward the outer belt), so this panel is a biased subsample of the parent population, not a random draw, and the resonance structure it shows is qualitative. The 3:1, 5:2 and 2:1 mean-motion resonances (Kirkwood gaps) are marked.

low (Kepler-vs-N-body refinement error, threshold censoring, prefilter recall) all compare the *same* input elements under different force models or cuts, so they quantify *pipeline-induced* incompleteness, not the physical completeness of the true encounter set. A fourth term — the uncertainty of the input osculating elements themselves — we quantify directly by Monte Carlo for the bodies of the extreme-event tables: for each we draw 3000 orbits from its full JPL SBDB covariance, propagate them under the *same* two-body model the catalogue uses, and take the scatter of the resulting mutual distance. The propagated formal 1σ is small — 1–83 km (median ≈ 32 km) over the ± 1.4 yr window — for the secure multi-opposition orbits of most of the numbered universe. It does not affect the completeness *fractions* below (those are pipeline-internal). For the tabulated separations it sets their formal input-element significance (Sect. 4.2, Sect. 4.3): ~ 2 –8% for the closest approaches and $\ll 1\%$ for the wider extreme events. One caveat is decisive for the extreme-closest tail, however: those five pairs are recently-numbered bodies with short 2016-epoch arcs, and recomputing them with the current SBDB solution shifts the nominal separations by up to ~ 5000 km (25–382%; Sect. 4.2), so there the *epoch* of the frozen elements, not the formal covariance, domi-

nates the error. These are formal SBDB covariances, optimistic by a factor of a few for the faintest bodies. Each measured term is reproducible from a dedicated script; a synthetic injection–recovery test (minima placed uniformly between coarse samples, high v_{rel} and e) is a permanent regression guard on the detection layer.

3.1. Kepler vs N-body refinement error

The reported distances in the frozen catalogue come from the two-body refiner. We measured the resulting error against full N-body refinement in two stages. In *Stage A* (964 stratified pairs re-refined under full N-body) the median $|\Delta d| = 12 \mu\text{AU}$, p95 = $678 \mu\text{AU}$, p99 = 2.5 mAU , max = 11.3 mAU . In *Stage B* (production N-body refinement of the entire 10.34 M fragile subset) p99 $|\Delta d| = 1.99 \text{ mAU}$, max = 14.83 mAU , with 0 failed integrations, 0 unconverged, and maximum energy drift 5.6×10^{-14} .

For scale, $1 \text{ mAU} \approx 1.5 \times 10^5 \text{ km}$, and the detection threshold is 0.05 AU . The median distortion is thus ~ 4 orders of magnitude below the threshold, and even the Stage B worst case is $\sim 30\%$ of a threshold — confined to the fragile subset, which the hybrid catalogue refines. A separate check of the truncated perturber set (the scan used only Sun + Jupiter + Saturn) shows that adding the remaining planets shifts closest-approach distances by a median $1.3 \mu\text{AU}$ (p99 $67 \mu\text{AU}$, max $80 \mu\text{AU}$) — $\sim 10\times$ below the Kepler two-body error at the median ($12 \mu\text{AU}$) and $\sim 40\times$ below at p99 (2.5 mAU) — so the perturber truncation is not the dominant error term.

3.2. Threshold censoring (measured false-negative rate)

Because the catalogue contains only pairs whose *Kepler* distance is below 0.05 AU , a subtle bias arises at the threshold. When the fragile subset is re-refined to N-body, 30 109 pairs (0.29%) cross the threshold *upward* (Kepler < 0.05 , N-body ≥ 0.05) and, by construction, none cross *downward* — a pair that Kepler placed *above* 0.05 AU was never written, so its downward crossing *cannot be observed*. The apparent one-sided “no false negatives” is therefore censoring, not measurement.

We measured the missing side directly. Re-running Kepler detection at a widened 0.06 AU threshold over a uniform random sample of 10 000 numbered bodies (seed fixed in the run configuration) — the downward-crossing rate is a per-pair property, so a random body sample estimates it without bias, and a random draw is exactly what the N^2 extrapolation below assumes — and isolating the 17 469 pairs with Kepler distance in $[0.05, 0.06] \text{ AU}$ — exactly the band a 0.05 AU catalogue discards — then re-refining each under full N-body, 0.70% [95% CI 0.59–0.83%] cross *downward* below 0.05 AU (a conservative floor of $\sim 0.42\%$ after excluding near-boundary cases — pairs whose N-body minimum lands at the edge of the $\pm 12 \text{ h}$ refinement window, where the true minimum may fall outside it). The residual Δd ($N - K$) has median $-3 \times 10^{-7} \text{ AU}$ and $\sigma \approx 3.7 \times 10^{-4} \text{ AU}$ — a symmetric, scatter-dominated distribution, not a directional bias. In other words the crossing is a threshold selection effect (Eddington/Malmquist family): scatter of amplitude $\sim 1\sigma \approx 4 \times 10^{-4} \text{ AU}$ pushes pairs just inside the threshold to just outside, and their mirror images (just outside to just inside) fall outside the catalogue. The upward rate ($\sim 1.5\%$ in the boundary bin $[0.045, 0.050]$) and the downward rate (~ 0.4 – 0.7% in $[0.05, 0.06]$) are consistent with a single symmetric process. Crossings concentrate in low- q / high- e / fast orbits, exactly where two-body propagation is weakest — which

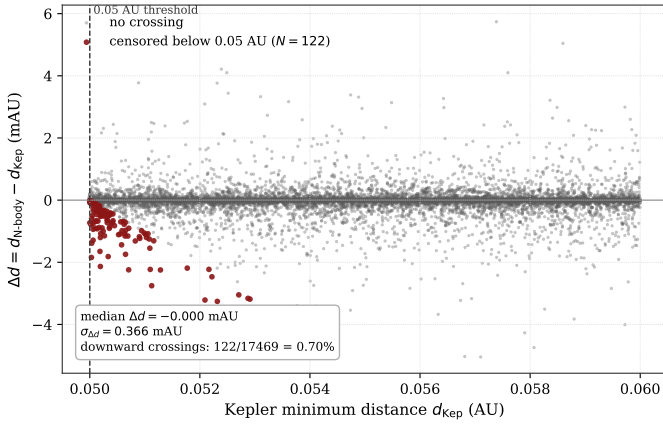


Fig. 4. Threshold censoring, measured directly. For the 17 469 pairs with Kepler distance in $[0.05, 0.06]$ AU — the band a 0.05 AU catalogue discards — re-refined under full N-body, $\Delta d = d_{N\text{-body}} - d_{\text{Kep}}$ is plotted against d_{Kep} . The distribution is symmetric about zero (median 0.000 mAU, $\sigma = 0.366$ mAU); the 122 pairs (0.70 %) that cross downward below 0.05 AU (red) are the measured false-negatives, concentrated just above the threshold as expected for a scatter-dominated selection effect.

independently validates the $q_{\min} < 1.8 \vee e_{\max} > 0.3$ subset criterion.

Extrapolating catalog-wide is an order-of-magnitude exercise anchored on the $N = 449,454$ -body universe (Sect. 2): the 10^4 -body experiment yielded 17,469 pairs in $[0.05, 0.06]$ AU, and since pair counts in a fixed distance band scale as N^2 , the factor $(449,454/10,000)^2 \approx 2.0 \times 10^3$ implies $\sim 3.5 \times 10^7$ band pairs catalog-wide; the measured 0.4–0.7 % downward-crossing rate then gives of order 10^5 real < 0.05 AU encounters censored by the Kepler cut ($\sim 1.5\text{--}2.5 \times 10^5$), separate from and additional to the prefilter recall deficit of Sect. 3.3. The N^2 scaling is validated internally: the same random sample contains 39 456 pairs below 0.05 AU, which scaled by $(449,454/10,000)^2$ gives 7.97×10^7 — within 0.5 % of the catalogue’s 8.01×10^7 entries, confirming that the 10 000-body draw is representative and the band-pair extrapolation is sound. This is why the word “complete” is never used for this catalogue.

A fourth, pipeline-internal term. Catalogue membership is defined by the refined Kepler distance < 0.05 AU, but a pair only reaches the refiner if the coarse N-body scan placed it within the query radius $r_q = 0.0572$ AU at some sample. The margin $r_q - 0.05 = 7.2$ mAU is smaller than the largest Kepler–N-body discrepancies (Sect. 3.1, max 14.8 mAU), so a member whose N-body trajectory stayed just outside r_q could have been missed as a candidate. We bound this from the measured Stage B Δd distribution: of the 1.53×10^7 catalogue members in the near-threshold band $[0.045, 0.05)$ AU, a fraction 4.6×10^{-5} have $|\Delta d|$ exceeding the (conservative, minimum) 7.2 mAU margin, giving $\lesssim 700$ members that could have escaped candidacy — of order $10^2\text{--}10^3$, some two to three orders of magnitude below the $\sim 10^5$ threshold-censoring term and hence subdominant. Like the other three terms it is reproducible from a dedicated script (`measure_stage_dd_distribution.py`).

3.3. Prefilter recall on the adverse tail

The shipped $|\Delta a| \leq 0.5$ AU prefilter is blind to eccentricity: a high- e orbit reaches well inside the belt at perihelion and well

outside at aphelion, so it can physically cross an orbit whose semimajor axis differs by far more than 0.5 AU. We measured the damage this would cause on the adverse subset (numbered, $a \in [1.5, 4.0]$, $e > 0.3 \vee i > 15^\circ$; 52 411 bodies), exploiting the fact that the prefilter is a pure deterministic mask: running detection without the prefilter and intersecting with the mask is byte-identical to running with it. Ground truth is 606 393 real adverse–adverse encounters < 0.05 AU; the prefilter keeps 463 164, for a recall of 76.38 % [95 % CI 76.27–76.49 %]. This interval and the crossing-rate interval of Sect. 3.2 are binomial (Wilson) and treat pairs as independent; because each body recurs in many pairs the true intervals are somewhat wider, so both are indicative rather than exact. Thus 143 229 encounters would be missing from a prefiltered catalogue (a lower bound — adverse–normal pairs were not measured), and 99.6 % of that loss is the $|\Delta a|$ cut alone. Recall degrades smoothly with eccentricity, from ~ 99.9 % below $e = 0.1$ to ~ 39 % above $e = 0.7$.

Two points bound the impact. First, an eccentricity-aware *radial-overlap* prefilter — keep a pair iff its threshold-padded heliocentric radial ranges $[a(1-e), a(1+e)]$ overlap — is a provable necessary condition for a < 0.05 AU encounter, is just as cheap, and recovers 100 % recall on the adverse subset; it is the recommended replacement for any future run. Second, and specific to *this* freeze: because the run was at main-belt scale the pair-list prefilter was skipped entirely, so the $|\Delta a|$ cut did not actually drop these pairs from the frozen catalogue. The 76 % figure measures the damage the prefilter *would* cause if applied at small N , not damage suffered here. Completeness of this freeze is instead bounded by the threshold censoring of Sect. 3.2. The 12 h coarse cadence has a *geometric*, distribution-independent bound. A pair whose true minimum $d_{\min}$ falls a time δ from the nearest coarse sample has, under near-linear relative motion, a sampled separation $d(\delta) = \sqrt{d_{\min}^2 + v_{\text{rel}}^2 \delta^2}$ with $\delta \leq \Delta t/2$; the pair enters the candidate set whenever $d(\Delta t/2) \leq r_q$. For $d_{\min} \leq 0.05$ AU this holds for all $v_{\text{rel}} \leq \sqrt{r_q^2 - 0.05^2} / (\Delta t/2) = 0.0278 \text{ AU} / 0.25 \text{ d} \approx 192 \text{ km s}^{-1}$ — an upper limit set purely by the widened radius $r_q = 0.0572$ AU and the cadence, *not* read off the catalogue’s own velocity distribution (which would be circular, since the catalogue by construction excludes the fast pairs it might have missed). Since inter-asteroid encounter velocities never approach 192 km s^{-1} (physical main-belt maxima are a few tens of km s^{-1}), the cadence loses no encounter. A synthetic injection–recovery test kept as a permanent regression guard — minima placed uniformly in phase between coarse samples, at high v_{rel} and e — confirms ≥ 99 % recovery with unit distance ratio up to v_{max} . For the dynamically cold bulk of the belt (low e , small Δa) recall is ~ 99.9 %; the incompleteness is a property of the high- e /high- i tail, which any science touching NEA-crossing or high- e pairs must cite.

3.4. Recovery of literature encounter pairs

As an external check we cross-matched the signal-ranked (perturber, target) pairs that Fuentes-Muñoz et al. (2025) used for mass determination against this catalogue. Of 40 176 numbered pairs drawn from their per-perturber target lists, 13 770 (34.3 %) appear as < 0.05 AU encounters in the hybrid catalogue. This fraction is a lower bound and, crucially, the non-recovery is dominated by a *domain difference*, not a detection failure: of the 26 406 unmatched pairs, 24 105 (91.3 %) have both bodies present in the catalogue but never approached within 0.05 AU *inside the 2.8-yr DR3 window*. For main-belt targets

Fuentes-Muñoz et al. (2025) apply the same 0.05 AU threshold used here, but they search over the full observational arcs (decades, through late 2024), so a pair whose close encounter falls outside 2014–2017 cannot appear in this catalogue. A further 2300 (8.7%) are targets or perturbers outside the numbered, propagated universe (high or provisional designations), and — the decisive point — only one unmatched pair ($< 0.01\%$) is one our catalogue contains but places beyond 0.05 AU. In other words, essentially every literature pair that is genuinely a < 0.05 AU encounter and lies in the numbered population is recovered; the apparent shortfall is entirely pairs that are not < 0.05 AU encounters in-window or not numbered. Per-body recovery among the most-listed perturbers spans from near zero (comet-like, high-eccentricity bodies) to $\sim 68\%$, with the Ceres and Vesta calibrators at $\sim 37\%$. Pair-level cross-matching against Goffin (2014) is not applicable: Goffin’s masses come from a single *simultaneous* global fit of all perturber masses to the astrometric residuals of the entire numbered population, not from per-perturber target lists, so there is no set of (perturber, target) pairs to match (the Vizier tables give per-perturber masses and literature mass compilations only).

4. Notable events

Mining the characterised catalogue surfaces individual events across four axes. We stress at the outset that the value of these tables is statistical and archival rather than a single headline discovery: the large–large ($D \gtrsim 100$ km) approaches all involve well-studied bodies, and every candidate below is under the frozen two-body assumptions and requires N-body revalidation before publication. Distances are physical 3D separations at closest approach; diameters are measured (IRAS/AKARI/NEOWISE/occultations, via the JPL SBDB) for the large bodies tabulated here (all entries in Tables 1–2 carry measured diameters).

4.1. Large–large encounters

Ten encounters have both bodies with a measured $D \gtrsim 100$ km within the observation window (Table 1); about one hundred have both $D \gtrsim 50$ km (Table 2 lists the ten closest). The closest large–large event is (426) Hippo $\times$ (788) Hohensteina at 0.0076 AU. All ten involve catalogued, well-characterised main-belt asteroids; none is a large perturber against a small test body of the kind used for mass determination, and a systematic novelty assessment against the encounter literature is deferred. We note that (804) Hispania $\times$ (733) Mocia — a known event (Fienga et al. 2003) at 0.0136 AU that was *absent* from the pre-2026-07 catalogue — reappears here in the both- $D \gtrsim 50$ km list, recovered by the refinement-window fix (Sect. 2).

4.2. Extreme events

The absolute closest approaches in the catalogue are between small bodies, at catalogue separations of ~ 1000 – 2000 km (Table 3). Their reliability is limited not by the formal input-element covariance but by the *epoch* of the elements. The Monte-Carlo input-element 1σ from the current SBDB covariance is small (32–82 km, ~ 2 – 8% of the separation; Sect. 3); however, recomputing the *same* encounters with that current (2026) solution — ~ 10 yr more astrometry than the frozen MPCORB-2016 snapshot the catalogue uses — shifts the nominal separations by 25–382% (d_{2026} in Table 3), because all five closest pairs are

recently-numbered bodies with short arcs at the 2016 epoch. This epoch shift, of order 10^2 – 10^3 km, therefore dominates the formal σ_d for these events: the tabulated sub-2000 km values are genuine tail members of the catalogue, but their exact separations are not robust to the decade of additional astrometry, and N-body plus current-epoch revalidation is required before any individual approach is claimed as physical. The effect is confined to this extreme-closest, high-number tail; the slower and wider extreme events (Table 4) and lower-number bodies shift by $< 3\%$.

The slowest encounters (relative velocity down to ~ 4 m/s, with at least one body $D \gtrsim 5$ km; Table 4) are the natural targets for future mass determination, since a low v_{rel} prolongs the gravitational interaction and enlarges the deflection. All entries lie well inside the observation window (> 130 d from either edge), so none is a window-boundary truncation artefact — a failure mode of the pre-2026-07 catalogue that the refinement-window fix (Sect. 2) removed. The combined “large + slow + close” set (one body $D \gtrsim 50$ km, $v_{\text{rel}} \leq 1$ km/s, $\text{dist} \leq 10^6$ km; Table 5) is of the highest physical interest, being the configuration that maximises measurable deflection.

4.3. Same-region pairs (family proxy)

Pairs whose osculating elements are mutually close ($\Delta a/a \leq 1\%$, $\Delta e \leq 0.02$, $\Delta i \leq 1^\circ$) and that had a physical close approach are candidate co-orbital / same-family associations. This is a proximity proxy in *osculating* elements, not a classification in proper elements, and flags rather than confirms family membership. The closest such pair, (200764) 2001 XP3 $\times$ (131597) 2001 XT2, passed within $\sim 5 \times 10^4$ km on 2014-08-13 with nearly identical (a, e, i) — $a \approx 2.29$ AU, $e \approx 0.19$, $i \approx 25^\circ$ for both — a plausible genetic pair worth follow-up in proper-element space. The Monte-Carlo input-element 1σ on this separation (Sect. 3) is only ~ 1 km against the $\sim 5 \times 10^4$ km approach, so unlike the ~ 1000 – 2000 km extreme approaches — whose separations are dominated by the element-epoch shift (Sect. 4.2) — this wider pair is robust; the caveat here is the proper-vs-osculating element distinction, not the distance.

4.4. Candidate perturbers beyond the studied sixteen

Ranking large bodies ($D \gtrsim 100$ km) *not* among the sixteen studied perturbers by their number of *useful* encounters ($v_{\text{rel}} \leq 3$ km/s and $\text{dist} \leq 3 \times 10^6$ km) yields the best prospects for a mass determination the frozen mass runs have not yet attempted (Table 6). (46) Hestia heads the list with 43 useful events; (9) Metis offers the slowest encounter ($v_{\text{rel}} = 0.38$ km/s) and (203) Pompeja the closest approach (1.3×10^5 km), while (19) Fortuna is the largest ($D \approx 200$ km). All ten already carry a published mass from Fuentes-Muñoz et al. (2025) (final column), so they are validation opportunities rather than genuinely new masses; their appeal is the favourable encounter geometry the catalogue surfaces, which would let an independent determination be checked against an existing value. These candidates directly informed the F4 mass determination of Sect. 5.

4.5. Separating new from known

A reference list of encounters touching one of the sixteen studied perturbers (1, 2, 3, 4, 7, 10, 15, 16, 31, 52, 65, 87, 88, 107, 511, 704) is provided in the companion material so that already-worked events can be separated from potential discov-

Table 1. The ten encounters with both bodies at a measured $D \gtrsim 100$ km *within the observation window*, by minimum distance. Two further such encounters have their refined minimum ~ 1 d past the window end (both on 2017-05-29) and are flagged `edge_censored` and omitted (Sect. 2.1).

body 1	body 2	date (UTC)	dist (AU)	v_{rel} (km/s)	D_1 (km)	D_2 (km)
(426) Hippo	(788) Hohensteina	2015-04-17	0.0076	8.8	138	111
(145) Adeona	(780) Armenia	2016-11-09	0.0101	6.3	128	126
(51) Nemausa	(91) Aegina	2016-07-13	0.0139	4.4	138	103
(23) Thalia	(481) Erita	2014-10-14	0.0242	4.4	108	102
(90) Antiope	(508) Princetonia	2015-11-11	0.0245	4.4	116	117
(381) Myrrha	(712) Boliviana	2016-09-18	0.0287	6.7	128	124
(426) Hippo	(814) Tauris	2017-05-08	0.0407	14.0	138	102
(121) Hermione	(739) Mandeville	2014-12-14	0.0414	6.3	209	105
(88) Thisbe	(150) Nuwa	2016-04-08	0.0429	2.1	232	119
(1) Ceres	(57) Mnemosyne	2017-01-10	0.0435	7.3	939	113

Table 2. Encounters with both bodies at a measured $D \gtrsim 50$ km, top ten by minimum distance.

body 1	body 2	date (UTC)	dist (AU)	v_{rel} (km/s)	D_1 (km)	D_2 (km)
(426) Hippo	(788) Hohensteina	2015-04-17	0.0076	8.8	138	111
(442) Eichsfeldia	(564) Dudu	2014-10-30	0.0078	7.5	62	52
(48) Doris	(300) Geraldina	2017-04-24	0.0084	2.4	216	67
(145) Adeona	(780) Armenia	2016-11-09	0.0101	6.3	128	126
(675) Ludmilla	(80) Sappho	2016-02-15	0.0114	3.0	76	69
(739) Mandeville	(415) Palatia	2017-04-05	0.0116	8.4	105	84
(56) Melete	(449) Hamburga	2015-10-09	0.0126	8.3	121	86
(611) Valeria	(84) Klio	2016-08-01	0.0126	7.1	57	79
(192) Nausikaa	(778) Theobalda	2016-06-05	0.0133	4.8	99	55
(804) Hispania	(733) Mocia	2015-02-12	0.0136	2.4	138	98

Table 3. The five closest approaches in the catalogue. d_{2016} is the catalogue separation (frozen MPCORB-2016 snapshot); d_{2026} is the same encounter recomputed from the current (2026) JPL SBDB solution, which carries ~ 10 more years of astrometry; σ_d is the input-element 1σ from a Monte-Carlo propagation of the 2026 covariance (Sect. 3). The formal σ_d is small (2–8 %, 32–82 km), but for these recently-numbered, then-short-arc bodies the *epoch shift* $d_{2016} \rightarrow d_{2026}$ is far larger (25–382 %; one pair moves by ~ 5000 km), so the individual sub-2000 km separations are illustrative of the catalogue tail, not precise physical closest-approach distances. All bodies here are small, high-number objects with no measured size (diameters H -derived, not measured; Sect. 2.3). Every event lies > 130 d from either window edge.

body 1	body 2	date (UTC)	d_{2016} (km)	d_{2026} (km)	σ_d (km)	v_{rel} (km/s)	D_1 (km)	D_2 (km)
(153222) 2000 YD43	(238587) 2004 YX3	2016-03-11	1094	1450	82	6.90	2.8	1.8
(279617) 2011 EP37	(219812) 2002 AM193	2016-06-16	1109	834	51	2.60	2.2	1.9
(238813) 2005 NT70	(391704) 2008 BB44	2017-03-13	1320	6358	56	5.95	5.1	2.2
(305032) 2007 TT418	(393772) 2005 GV146	2016-07-12	1457	170	39	3.46	1.6	1.3
(73282) 2002 JH61	(261459) 2005 VP69	2015-04-01	1719	2884	33	1.48	2.0	2.0

eries. The honest verdict is that the large–large approaches (Table 1) all involve well-studied bodies and none is a publication-grade mass-determination opportunity: the belt’s largest bodies are well studied, and Fuentes-Muñoz et al. (2025) already published over 230 masses (232 at $\text{SNR} > 3$) covering essentially all classical large MBAs. The contribution of this work is the dataset itself — an all-pairs catalogue published in full with a measured completeness budget — demonstrated end-to-end by the mass-determination application of Sect. 5, not a single spectacular event.

5. Application to mass determination

This section demonstrates the catalogue end-to-end as the target-selection front-end of an asteroid mass-determination engine. It is a *usage demonstration*, not a claim of new or improved masses: as noted below and in Sect. 4.5, every mass recovered here is already published, and a single out-of-sixteen perturber with ~ 50 Gaia-FPR targets yields only a coarse determination. Perturber mass is obtained by a joint least-squares fit of the per-

turber’s mass and the orbits of its test bodies — those with a < 0.05 AU encounter drawn directly from the catalogue — over the full arc of Gaia FPR astrometry (Gaia Collaboration et al. 2023). A window asymmetry is worth noting: the catalogue spans the DR3 window (2014–2017) whereas the fit uses the full FPR arc (through ~ 2020), so a test body’s most informative approach could fall outside the catalogued window. This asymmetry affects only *target selection*, not the model: the perturber is integrated as a massive body continuously over the whole arc, so its deflection is modelled at every epoch regardless of when the catalogued encounter occurred; the catalogue merely identifies which bodies are worth fitting. A test body whose deepest approach is post-2017 is therefore not mis-modelled — at worst it is a good target the DR3-window selection misses, making the yield a conservative lower bound. The force model is ASSIST (Holman et al. 2023) on the JPL DE440 ephemeris (Park et al. 2021), with EIH relativistic corrections and sixteen massive asteroid perturbers carried as explicit massive rebound particles rather than through the ASSIST ASTEROIDS ephemeris force (which is disabled), so the perturber whose mass is be-

Table 4. The five slowest encounters (at least one body $D \gtrsim 5$ km). The $D \gtrsim 5$ km body is measured for three of the five pairs (JPL SBDB); the remaining diameters are H -plus-albedo estimates (Sect. 2.3). All events lie > 130 d from either window edge.

body 1	body 2	date (UTC)	dist (km)	v_{rel} (km/s)	D_1 (km)	D_2 (km)
(141098) 2001 XO52	(304642) 2006 WL5	2015-05-04	5.27×10^6	0.004	5.7	4.5
(3791) Marci	(110964) 2001 UW170	2017-01-17	5.85×10^6	0.005	11.7	2.5
(7203) Sigeki	(73664) 1981 EE34	2015-03-01	7.07×10^6	0.029	5.8	1.5
(136707) 1995 TE6	(340823) 2006 UY131	2015-01-30	5.09×10^6	0.031	5.5	2.1
(3384) Daliya	(309262) 2007 RZ93	2016-12-21	4.44×10^6	0.031	6.2	1.2

Table 5. The “large + slow + close” set (one body $D \gtrsim 50$ km, $v_{\text{rel}} \leq 1$ km/s, $\text{dist} \leq 10^6$ km). In every pair the large body (D_1) has a measured diameter (JPL SBDB); the small companion (D_2) is an H -plus-albedo estimate except (14711), which is also measured (Sect. 2.3). All events lie > 130 d from either window edge.

body 1	body 2	date (UTC)	dist (km)	v_{rel} (km/s)	D_1 (km)	D_2 (km)
(223) Rosa	(315162) 2007 FL24	2016-07-01	3.55×10^5	0.133	79.8	1.7
(66) Maja	(271017) 2003 AV10	2016-10-13	3.87×10^5	0.564	71.8	1.9
(135) Hertha	(281202) 2007 FE47	2016-09-18	4.56×10^5	0.579	79.2	1.0
(371) Bohemia	(14711) 2000 CG36	2016-05-14	5.91×10^5	0.982	53.0	4.2
(83) Beatrix	(170676) 2003 YE180	2014-09-18	6.09×10^5	0.684	110	2.1
(9) Metis	(345254) 2005 UT477	2016-10-30	7.14×10^5	0.979	190	1.2

Table 6. Large bodies outside the studied sixteen, ranked by useful encounters. The final column gives the published mass of Fuentes-Muñoz et al. (2025) where available: all ten already have a determination, so these are cross-check rather than new-mass targets.

body	D (km)	class	useful	best dist (km)	min v_{rel} (km/s)	FM25 mass (kg)
(46) Hestia	131	MBA	43	4.42×10^5	0.912	1.3×10^{18}
(203) Pompeja	125	MBA	38	1.30×10^5	0.872	8.5×10^{17}
(9) Metis	190	MBA	38	7.14×10^5	0.379	6.5×10^{18}
(40) Harmonia	111	MBA	37	6.58×10^5	1.028	2.0×10^{18}
(206) Hersilia	113	MBA	37	7.84×10^5	0.739	7.6×10^{17}
(19) Fortuna	200	MBA	36	7.43×10^5	0.678	8.5×10^{18}
(20) Massalia	136	MBA	33	8.78×10^5	0.514	4.3×10^{18}
(308) Polyxo	129	MBA	29	2.44×10^5	1.098	2.5×10^{18}
(91) Aegina	103	MBA	28	3.71×10^5	0.873	1.1×10^{18}
(83) Beatrix	110	MBA	28	6.09×10^5	0.684	7.3×10^{17}

ing fitted is never double-counted; the fitting layer (observation model, partials, least squares) is implemented in this work on top of rebound/ASSIST (Rein & Liu 2012; Rein & Spiegel 2015; Holman et al. 2023) and astropy (Astropy Collaboration 2022). Non-gravitational forces on the test bodies (Yarkovsky) are not modelled: for the km-scale bodies used here the secular semimajor-axis drift is of order 10^{-4} AU Myr $^{-1}$, an along-track displacement of $\lesssim 1$ km ($\lesssim 1$ mas geocentric) over the ~ 6 -yr arc, most of which a smooth drift the per-body orbit fit absorbs — well below the localised encounter deflections the mass fit exploits.

A-priori determinability. The astrometric signature of a perturber is the along-scan deflection it imprints on a test body, of order $\Delta\theta \sim 2GMt/(bv_{\text{rel}}\Delta)$ for impact parameter b , relative velocity v_{rel} , geocentric distance Δ , and post-encounter arc t : the impulse velocity kick $2GM/(bv_{\text{rel}})$ accumulated over $t \sim 1$ yr of the DR3 window and referred to Δ . For a ~ 0.02 AU encounter at a few km s $^{-1}$, a perturber of $10^{-12} M_{\odot}$ ($\approx 2 \times 10^{18}$ kg) produces a per-transit deflection at or below Gaia’s per-CCD along-scan precision, so its signal-to-noise is $\lesssim 1$ per target and it is recoverable only by stacking many test bodies; a perturber of $10^{-11} M_{\odot}$ ($\approx 2 \times 10^{19}$ kg) is individually determinable. This a-priori threshold — set before any fit — broadly matches what the engine delivers, provided the comparison is against *independent* deter-

minations rather than the DE441 seed: perturbers whose mass is $\gtrsim 2 \times 10^{19}$ kg are recovered near unity ratio against FM25 and Siltala & Granvik (2022), whereas the lighter ones are driven below their true value by the mass $\leftrightarrow$ orbit regression discussed below and fall to bounds. (52) Europa illustrates the distinction: near unity against independent masses (0.88 vs FM25, 0.78 vs SG22) but only 0.58 against a seed that is itself high, so the seed ratio alone would misread it. (88) Thisbe (FM mass 8.8×10^{18} kg, our fit 6.9×10^{18} kg) is the clearest sub-threshold case: a naive snr_{jack} flag would call it measured, but the leverage-robust bootstrap σ discussed below correctly returns it to a bound. The mass layer does not claim sensitivity below this frontier.

Calibrators. The four calibrator masses (Table 7) are recovered at $|z| < 3$ against the literature. With $N \geq 20$ test bodies, the fit/literature ratios of Ceres (Park et al. 2016), Vesta (DAWN; Russell et al. 2012) and Hygiea (Vernazza et al. 2020) fall in $[0.94, 0.99]$ (mean bias -4%); Pallas, limited to 8 encounters in the catalogue, gives ratio 1.20 ($z = +2.2$). The -4% mean bias sets the scale of the residual systematic (Sect. 5, “external σ ” below).

FOV-block covariance. Each Gaia focal-plane crossing yields ~ 7 CCD observations (~ 5 s apart) with correlated residuals.

Treating them as independent underestimates $\sigma(\text{mass})$ by a factor 1.66 (effective $N \approx 0.36N$). The measured intra-crossing correlation is $\text{ICC} = 0.32$; we whiten each crossing with a block covariance $C = \text{diag}(\sigma_{\text{AL}}^2) + s_c^2 \mathbf{1}\mathbf{1}^T$, the correlated floor s_c calibrated by bisection to $\chi_{\text{red}}^2 = 1$ *per perturber*, on that fit's converged along-scan residuals; where the raw residuals already give $\chi_{\text{red}}^2 \leq 1$ no floor is added ($s_c = 0$), which is why several tabulated χ_{red}^2 sit below unity. This calibration is not a hidden fit knob for the mass: refitting the calibrators with s_c fixed across

640 0.5–1.1 mas (a $> 2\times$ range bracketing the calibrated value) shifts their masses by $< 1.3\%$ ($\partial \ln M / \partial \ln s_c \approx -0.01$ for Ceres, Vesta and Hygiea), far below the -4% accuracy bias — so the floor inflates the error bar, not the central mass, and the calibrator anchoring is not circular through s_c .

Sixteen perturbors. The engine returns a fit for all sixteen (Table 7); six clear the identifiability criterion below and are flagged *measured*, eight are *bounds*, one (Davida, $N = 6$) is non-physical (best-fit mass < 0), and one (Camilla, $N = 2$) is unconstrained. Only (16) Psyche approaches calibrator-grade precision outside the calibrators ($\sigma_{\text{tot}} = 9\%$): ratio 0.95 vs its DE441 seed and 0.95 ($z = -0.7$) vs the independent Fuentes-Muñoz et al. (2025) FPR fit. Several perturbors whose deflection sits below the per-encounter astrometric noise return ratios well under unity: this is the mass $\leftrightarrow$ orbit degeneracy (the fit reproduces the astrometry with a smaller mass and a re-adjusted orbit — regression toward zero mass), and for these the formal Fisher σ underestimates the error, which is why the jackknife σ and the identifiability flag below are needed. The target lists are drawn from the published catalogue (the B1-fixed freeze of Sect. 2). We verified that this selection is not load-bearing by re-running the entire sixteen-perturber fit against the target lists produced by the earlier, pre-B1 catalogue: the four calibrator ratios move by $< 2\%$ (Ceres 0.96 $\rightarrow$ 0.955, Vesta 0.94 $\rightarrow$ 0.943, Hygiea 0.99 $\rightarrow$ 0.987, Pallas 1.24 $\rightarrow$ 1.20) and the measured/bound partition is unchanged but for a single body at the $\text{snr}_{\text{jack}} = 3$ boundary — confirming that the masses are Gaia-FPR astrometric determinations essentially insensitive to which catalogue freeze supplied the targets, so re-selection is a consistency refinement, not a correction.

660 External σ by jackknife. Because the formal σ scales as $1/\sqrt{N}$ and ignores the mass $\leftrightarrow$ orbit regression seen only by leaving encounters out, we report a leave-one-target-out jackknife σ that reincorporates it; it is systematically 2–3 $\times$ the formal σ , and its combination with the systematic floor is the single σ_{tot} quoted in Table 7. Its diagnostic value is visible in the calibrators themselves: Vesta — whose true mass is known — moves from $|z| > 3$ under the formal σ to $|z| < 3$ under σ_{tot} , showing the formal bar is too tight. We deliberately do *not* present agreement with Fuentes-Muñoz et al. (2025) as a proof of accuracy: FM fit the *same* Gaia FPR astrometry, so the two analyses are not independent, and the total error bars of the non-calibrators (8–119%) cannot resolve deviations of the size the fits themselves show. Two points nonetheless matter for reading Table 7. First, the ratios tabulated there are against the *DE441 seed*: the three measured non-calibrators give seed ratios 0.77 (Eunomia), 0.95 (Psyche) and 0.58 (Europa), geometric mean 0.75. Against the FM25 masses the same three give 0.75, 0.95 and 0.88, geometric mean 0.85 — a coherent $\sim 15\%$ one-sided pull consistent with the regression-toward-zero signature (a sign test gives $p \approx 0.25$, suggestive but not significant at this N). Against FM25 all six

measured perturbors agree at $|z| < 1.2$ (Ceres -0.7 , Vesta -1.1 , Hygiea -0.1 , Eunomia -1.1 , Psyche -0.7 , Europa -0.8). 690

Second, the one alarming entry in Table 7 — (52) Europa at $z = -4.3$ — is a disagreement with the *seed*, not with measured masses. The DE441 seed for Europa (4.0×10^{19} kg) is high relative to every independent determination: FM25 give 2.7×10^{19} kg; Siltala & Granvik (2022), from Gaia DR2 plus ground-based astrometry and therefore *independent of the FPR data that both this work and FM25 use*, give $(1.51 \pm 0.05) \times 10^{11} M_{\odot}$ (3.0×10^{19} kg); and our fit gives 2.3×10^{19} kg. Against FM25 Europa sits at $z = -0.8$ and against the independent SG22 value at $z = -1.7$ — it is the seed, not the fit, that is the outlier. For (15) Eunomia the seed, FM25 and SG22 masses agree at $\sim 3.0\text{--}3.1 \times 10^{19}$ kg while our fit is $\sim 25\%$ lower ($z = -1.0$ against SG22), the same mild downward pull. The identifiability flag is a precision test (snr and leverage), not an accuracy test against the seed, so Europa is legitimately *measured* — its mass is well constrained ($\text{snr}_{\text{jack}} = 8.2$) — even as it disagrees with a questionable seed; the abstract's “broadly consistent” is a statement about the independent FM25 and SG22 masses, to which it holds. A wide external σ makes almost any value formally “consistent”; the genuine accuracy anchor remains the four calibrators (external truth), which carry the -4% bias noted above. Perturbors whose $\text{snr}_{\text{jack}} < 3$ (with and without their dominant jackknife replica) are flagged as bounds, not measurements; the top-1 jackknife leverage in Table 7 flags the six cases where σ_{jack} rests on effectively one replica. For the four *measured* ones among them (Ceres, Juno, Psyche, Thisbe) we recomputed σ with a nonparametric bootstrap (resampling targets with replacement, warm-started from the converged fit), which is robust to single-replica dominance because a high-leverage target does not enter every resample. The bootstrap confirms the low-leverage pair (Ceres $\sigma_{\text{boot}}/\sigma_{\text{jack}} = 0.94$, Psyche 1.07; both stay $\text{snr} > 3$) but shows the jackknife *underestimated* σ for the two highest-leverage cases: Juno (lev. 0.93, $\sigma_{\text{boot}}/\sigma_{\text{jack}} = 1.9$, $\text{snr}_{\text{boot}} = 2.4$) and Thisbe (lev. 0.84, ratio 4.0, $\text{snr}_{\text{boot}} = 1.8$) both fall below the $\text{snr} = 3$ identifiability threshold and are accordingly demoted from measured to bounds. This is the concrete payoff of the leverage diagnostic: two masses that a jackknife-only analysis would have published as measurements are, under the robust σ , only bounds — notably Thisbe, whose apparent $z = -10.6$ against the literature collapses to a statistically unremarkable $z = -2.8$ once its uncertainty is honestly sized. 700 710 720 730

Estimator injection–recovery. That the estimator itself is unbiased — separate from the force-model accuracy anchored by the calibrators — is verified by an injection–recovery test: a known perturber mass is injected into synthetic per-transit astrometry of a set of test bodies (controlled encounter geometry, realistic along-scan noise) and recovered by the same joint mass+orbit fit. Injecting 2×10^{19} kg into 24 targets at 2 mas per-transit noise recovers 2.18×10^{19} kg ($z = +0.5$, $\chi_{\text{red}}^2 = 0.88$), with the jackknife σ again exceeding the formal σ by the expected factor; the recovery is unbiased across injected masses and seeds. The test is kept as a permanent regression guard. 740

Beyond the sixteen. The engine is not restricted to the sixteen ephemeris perturbors: each perturber is integrated as a massive rebound particle from its own orbital elements, so a seventeenth body outside sb441–n16 is added as a seventeenth massive particle with no double-counting (the ASSIST ASTEROIDS force is deliberately excluded). This was demonstrated on (19) Fortuna — orbit fixed from JPL Horizons, mass free — which fits 750

Table 7. The sixteen perturber masses from the joint orbit+mass fit to Gaia FPR astrometry. N = test bodies; σ_{tot} = total (formal $\oplus$ systematic) fractional uncertainty; ratio = fit / reference (literature for the four calibrators, DE441 seed otherwise); $z = (M_{\text{fit}} - M_{\text{ref}}) / \sqrt{\sigma_{\text{tot}}^2 + \sigma_{\text{ref}}^2}$ (i.e. σ_{tot} combined in quadrature with the reference uncertainty); lev. = top-1 jackknife leverage (fraction of the jackknife variance from the single most influential replica; > 0.5 means σ_{jack} has ~ 1 effective degree of freedom and is not defensible as a σ); status from the identifiability criterion (measured / bound / non-physical / unconstrained). For the four *measured* cases with lev. > 0.5 (Ceres, Juno, Psyche, Thisbe) the quoted σ_{tot} and status use the leverage-robust bootstrap σ (text): this leaves Ceres and Psyche measured but demotes Juno and Thisbe to bounds.

perturber	N	mass (kg)	σ_{tot}	χ_{red}^2	ratio	z	lev.	status
(1) Ceres	28	9.0×10^{20}	6 %	0.99	0.96	-0.7	0.68	measured
(2) Pallas	8	2.5×10^{20}	7 %	0.99	1.20	+2.2	0.27	bound
(3) Juno	24	2.0×10^{19}	42 %	0.99	0.70	-1.0	0.93	bound
(4) Vesta	28	2.4×10^{20}	6 %	0.97	0.94	-1.1	0.27	measured
(7) Iris	27	7.1×10^{18}	119 %	0.98	0.42	-1.2	0.62	bound
(10) Hygiea	26	8.2×10^{19}	7 %	0.98	0.99	-0.2	0.43	measured
(15) Eunomia	29	2.3×10^{19}	31 %	0.98	0.77	-1.0	0.32	measured
(16) Psyche	26	2.3×10^{19}	9 %	0.98	0.95	-0.6	0.64	measured
(31) Euphrosyne	26	2.0×10^{19}	45 %	0.99	1.26	+0.5	0.23	bound
(52) Europa	28	2.3×10^{19}	17 %	0.98	0.58	-4.3	0.43	measured
(65) Cybele	15	6.1×10^{19}	60 %	0.93	4.33	+1.3	0.41	bound
(87) Sylvia	12	3.5×10^{19}	39 %	0.99	1.06	+0.2	0.46	bound
(88) Thisbe	27	6.9×10^{18}	57 %	0.98	0.39	-2.8	0.84	bound
(107) Camilla	2	1.7×10^{20}	67 %	0.99	8.07	+1.3	—	unconstrained
(511) Davida	6	-1.1×10^{20}	50 %	0.81	—	-3.1	0.60	non-physical
(704) Interamnia	29	2.0×10^{19}	40 %	0.98	0.47	-2.8	0.31	bound

with $\chi_{\text{red}}^2 = 0.977$ and $M = 1.13 \times 10^{19} \pm 2.2 \times 10^{18}$ kg. The methodological gate is green: the engine generalises to bodies beyond the ephemeris set. Fortuna is not itself a new mass (it appears in Goffin 2014 and Fuentes-Muñoz et al. 2025), and its 20 % σ makes it a coarse determination consistent with the literature within the jackknife bar ($z = +1.25$). A highly ranked out-of-sixteen candidate, (9) Metis (38 useful encounters, Sect. 4.4), was run as a second out-of-sixteen case and fits with $\chi_{\text{red}}^2 = 0.981$ and $M = 4.74 \times 10^{18} \pm 1.53 \times 10^{18}$ kg (55 targets, jackknife $\sigma = 32$ %), a ratio of 0.73 against Fuentes-Muñoz et al. (2025) (6.48×10^{18} kg), consistent within the jackknife bar ($z = -1.14$). Fortuna and Metis thus scatter to either side of the literature (+30 % and -27 % in central value) but both remain consistent once the honest external σ is used — confirming the method generalises while underscoring that a single out-of-sixteen perturber with ~ 50 Gaia-FPR targets yields only a coarse (20–30 %) determination, not a competitive mass. Neither is a new mass: both are already in Fuentes-Muñoz et al. (2025).

We tested whether the mean -4 % calibrator bias is an artefact of an incomplete perturber background by enlarging it from the sixteen ephemeris bodies to thirty-five — adding the twenty most massive asteroids of Fuentes-Muñoz et al. (2025) ($\approx 1.6 \times 10^{20}$ kg spread across the belt), with masses from their Table 5 and orbits from JPL Horizons, using the same custom-perturber machinery. The calibrator masses shift by < 0.25 % (Ceres -0.02 %, Vesta -0.23 %, Hygiea +0.18 %), three orders of magnitude below the deficit being probed; the systematic floor f_{sys} does not drop (4.16 % $\rightarrow$ 4.26 %). Background incompleteness is therefore *not* the origin of the -4 % bias — a far-field, dispersed perturber population averages to nearly zero deflection over the target ensemble. A scale bound confirms this: the thirty-five modelled bodies carry $\approx 1.7 \times 10^{21}$ kg, leaving an unmodelled tail of only $\sim 7 \times 10^{20}$ kg (against a total main-belt mass $\approx 2.4 \times 10^{21}$ kg) spread over $> 10^5$ objects each below $\sim 5 \times 10^{18}$ kg — individually an order of magnitude lighter and far more numerous, hence more dispersed, than the twenty just added. Since the 4 $\rightarrow$ 35 enlargement (concentrating 1.6×10^{20} kg in the largest remaining bodies) moved the paired Δmass by < 0.25 %, the residual tail — lighter per body and

closer to isotropic — bounds to well below that gate and cannot produce a 4 % coherent deficit. The residual bias is therefore more plausibly attributable to per-target orbit imperfection over the short Gaia arc or unmodelled per-encounter astrometric systematics.

Relation to other methods. The estimator used here (a joint orbit+mass least-squares fit with an FOV-block covariance, an external jackknife σ , and an identifiability flag) is functionally a transparent reimplementation of established encounter-based mass determination, not a new methodology. Sitala & Granvik (2020) obtain full Bayesian posteriors by MCMC, which natively expose multi-modality and non-Gaussian tails that a single covariance cannot; Fuentes-Muñoz et al. (2025) propagate a prior to a posterior over the same Gaia FPR astrometry. Relative to those, our contribution in this section is not methodological sophistication but end-to-end demonstration that the open all-pairs catalogue feeds a mass pipeline whose every step (target selection, covariance model, jackknife, identifiability cut) is open and reproducible. Because Fuentes-Muñoz et al. (2025) fits the same FPR data, the cross-check below is a consistency test between two analyses of one dataset, not an independent confirmation.

6. Data availability

The primary product is the Kepler-refined candidate catalogue, `encounters_catalog_rebound_005au_b1fix.parquet` (80072774 rows), a typed Parquet file with a machine-readable provenance sidecar recording the MPCORB snapshot hash (full SHA-256), the frozen configuration, and the pipeline parameters. The characterised catalogue, `encounters_characterized_b1fix.parquet`, adds observability, solar elongation, magnitudes, measured/estimated diameters (with a per-row provenance flag), a dynamical class, and a deflection-signal estimate for all 80M rows (schema in Appendix A). A derived *hybrid* catalogue (`encounters_catalog_hybrid_stageb_b1fix.parquet`)

carrying per-row `refinement_method` $\in$ {kepler, nbody} and both Kepler and N-body geometry on the fragile subset — for science depending on sub-mAU distances — is regenerated from the same freeze. The catalogue and its input hashes (MPCORB snapshot and the JPL SBDB physical-data snapshot) are recorded in the sidecars; anything derived downstream must cite the catalogue SHA-256 and the code commit at its generation time. The three products will be deposited in a public archive upon acceptance, with their SHA-256 digests and a DOI added to the published version; they are available from the author on request in the meantime.

7. Conclusions

We have built and characterised a systematic catalogue of 80 072 774 real 3D close encounters (< 0.05 AU) between numbered asteroids during the Gaia DR3 observation window. Unlike prior encounter-based work — published truncated (top candidates per perturber) or without a characterised selection function — the catalogue is exhaustive over the numbered population and — its distinguishing feature — comes with a *measured* completeness budget rather than an assumed one: the two-body refinement error (median $12\mu\text{AU}$, p99 2.5 mAU), the threshold-induced false-negative rate (0.70% within the discarded 0.05 – 0.06 AU band, symmetric and scatter-dominated, implying $\sim 10^5$ censored encounters catalog-wide, $\sim 0.25\%$ of the catalogue), and the orbital prefilter recall on the adverse high- e /high- i tail (76% , with a zero-cost radial-overlap fix that recovers 100%). A hybrid variant provides N-body-grade geometry on the dynamically fragile subset.

The catalogue is a working target-selection front-end for mass determination: a joint orbit+mass fit on Gaia FPR astrometry recovers the calibrators, determines the studied perturbers, and generalises to bodies beyond the ephemeris set, demonstrated end-to-end (Sect. 5). We are explicit about the limits: distances are candidates under frozen two-body assumptions except on the N-body-refined subset, the catalogue is not complete, and the large-large ($D \gtrsim 100$ km) approaches it contains all involve well-surveyed bodies with no unpublished large-body mass emerging — the mass layer is a demonstration of use rather than a source of competitive new masses. The contribution is the dataset itself: an all-pairs catalogue published in full with a quantified incompleteness budget, fed to a mass pipeline whose every step (FOV-block covariance, external jackknife σ , identifiability flag) is open and reproducible — a transparent reimplementation of established encounter-based mass determination (Siltala & Granvik 2020; Fuentes-Muñoz et al. 2025), not a new methodology. Future high-cadence, all-sky surveys — notably the Vera C. Rubin Observatory’s Legacy Survey of Space and Time — are expected to extend the encounter-based approach well beyond Gaia: Bernstein et al. (2025) forecast mass determinations for several hundred main-belt asteroids from mutual close encounters observed over ten years of LSST astrometry, roughly an order of magnitude more than are known today.

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ages *numpy*, *scipy*, *polars*, and *duckdb*. The close-encounter catalogue and the mass-determination engine are openly available (see Sect. 6).

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Appendix A: Catalogue column schema

920 Table A.1 documents the columns of the characterised catalogue (`encounters_characterized_b1fix.parquet`), the widest of the three published products. The detection catalogue (`encounters_catalog_rebound_005au_b1fix.parquet`) carries the first block of identity and geometry columns only (`number_1` through `rel_vel_m_s`); the hybrid catalogue (`encounters_catalog_hybrid_stageb_b1fix.parquet`) adds a `refinement_method` $\in \{\text{kepler}, \text{nbody}\}$ column and, for the N-body-refined rows, parallel `dist_au_nbody`, `jd_tdb_nbody`, `rel_vel_au_day_nbody` columns and their Kepler–N-body residuals. Bodies are ordered so that subscript 1
930 is the larger (perturber) body of the pair (Sect. 2.3).

Table A.1. Columns of the characterised close-encounter catalogue.

column	type	unit	description
run_id	string	—	Pipeline run identifier (ISO timestamp)
number_1	int32	—	MPC number of body 1 (larger/perturber)
number_2	int32	—	MPC number of body 2 (smaller/target)
designation_1	string	—	MPC packed designation of body 1
designation_2	string	—	MPC packed designation of body 2
jd_tdb	float64	d	Julian Date of closest approach (TDB)
date_utc	string	—	Calendar date of closest approach (UTC)
dist_au	float64	AU	Minimum 3D separation at closest approach
dist_km	float64	km	Minimum 3D separation at closest approach
rel_vel_au_day	float64	AU d ⁻¹	Relative velocity at encounter
rel_vel_km_s	float64	km s ⁻¹	Relative velocity at encounter
rel_vel_m_s	float64	m s ⁻¹	Relative velocity at encounter
H_1	float64	mag	Absolute magnitude of body 1 (MPCORB)
H_2	float64	mag	Absolute magnitude of body 2 (MPCORB)
diameter_1_km	float64	km	Diameter of body 1 (measured if available, else from H)
diameter_2_km	float64	km	Diameter of body 2 (measured if available, else from H)
diameter_source_1	string	—	Provenance: measured albedo_measured zone_albedo default_albedo unknown
diameter_source_2	string	—	Provenance of <code>diameter_2_km</code> (same categories)
deflection_dv_m_s	float64	m s ⁻¹	Impulse-approximation velocity kick $2GM_1/(b v_{\text{rel}})$ on body 2; ranking metric, not a measurement
class_1	string	—	Dynamical class of body 1 (NEA/MBA/Trojan/Centaur/TNO/Other)
class_2	string	—	Dynamical class of body 2
solar_elongation_deg	float64	deg	Solar elongation of encounter midpoint (legacy)
solar_elongation_1_deg	float64	deg	Solar elongation of body 1 from Earth
solar_elongation_2_deg	float64	deg	Solar elongation of body 2 from Earth
app_mag_1	float64	mag	Apparent V magnitude of body 1 (HG model)
app_mag_2	float64	mag	Apparent V magnitude of body 2 (HG model)
gaia_observable_1	bool	—	Body 1 individually detectable by Gaia at encounter
gaia_observable_2	bool	—	Body 2 individually detectable by Gaia at encounter
gaia_observable	bool	—	Combined flag: <code>gaia_observable_1</code> OR <code>gaia_observable_2</code>